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Characteristics and dynamics of electrolyzed bubbles in magnetohydrodynamic bubbly jet

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ABSTRACT

Background: The magnetohydrodynamics (MHD) are highly relevant to practical applications, including marine propulsion and hydrogen production etc. For better implementation to applications, typical characteristics of the MHD bubbly jet, such as the size distribution and motion of oxygen and hydrogen bubbles, need to be well realized.

Methods: The MHD bubbly jet generated by high current density associated with strong permanent magnet is experimentally investigated. Two fields of view are taken to analyze the details of micro-sized bubbles and the whole pattern of bubbly jet. To couple the magnetoelectric effects and bubble dynamics, the Reynolds number based on the Lorentz force Re_L is applied in the parametric analysis.

Findings: Dependence of the mean size of bubble on the current density is generally verified. Two regimes regarding the bubble sizes are distinguished. The proposed mechanisms of these two regimes are supported by the pattern of bubbly jet. For $Re_L \leq 2000$, size of bubble monotonically increases with current density, so that referred to as the electrochemistry regime. Nevertheless, the size of bubble remains nearly unchanged for $Re_L > 2000$, referred to as the flow regime in which the turbulent breakup of the bubbly jet core starts to evolve.

1. Introduction

Magnetohydrodynamic (MHD) is a unique branch of fluid dynamics dealing with conducting fluids driven by magnetoelectric effects of Lorentz force. Applications of MHD had been interests of engineers since its initiation in 1940s [1], including marine propulsion [2,3], pumps [4,5], motors [6], and silicon growth [7,8]. In the meantime, in the fields of enhanced heat transfer [9–13] and atmospheric pressure plasma jet [14,15], MHD is often applied in which the conductive medium is nanofluid and plasma, respectively.

More recently, because of the concern of greenhouse effect, production and utility of hydrogen energy is a truly important subject [16]. Hydrogen is typically produced by alkaline water electrolysis [17,18] because pure hydrogen can be generated by relatively simple equipment. Nevertheless, the yield rate of oxygen and hydrogen bubbles is low due to adhesion of the bubbles on the electrodes to reduce the reaction rate [19,20]. To improve the production rate of hydrogen, the Lorentz force of MHD has been successfully applied to enhance mass transfer and yield better efficiency [21–23]. Comprehensive reviews regarding the application of magnetic field to enhance hydrogen production can be referred to Ref. [24,25].

The characteristics of bubbles in electrolysis are also important issues studied intensively [26–32]. During electrolysis, the so-called bubble effect, i.e., adhesion and accumulation of bubbles near the electrodes, reduces the reaction rate. As a result, the covering area and dispersion speed of bubbles are important to increase hydrogen production [24]. In pure electrolysis, growth rate and the detached speed of big bubbles are vital in determining bubble size [27]. The bubble detachment diameter is inversely proportional to the flow velocity, but proportional to the current density [29]. After 10 seconds of electrolysis, diameter of oxygen bubble is about 55 μm to 170 μm for current density 0.5 A/cm^2 to 4 A/cm^2, respectively. Assisted by MHD convection, the bubbles rise faster to improve surface coverage of the electrode [26]. In addition to acceleration of bubble, the size of bubble increases with higher current density (or the Lorentz force) [28]. Removal of bubbles from the electrode plates decreases significantly by the presence of MHD effect [30]. The bubble size for current density of 2.3 A/cm^2 ~ 3 A/cm^2 after 0.4 second is around 250 μm ~ 300 μm.

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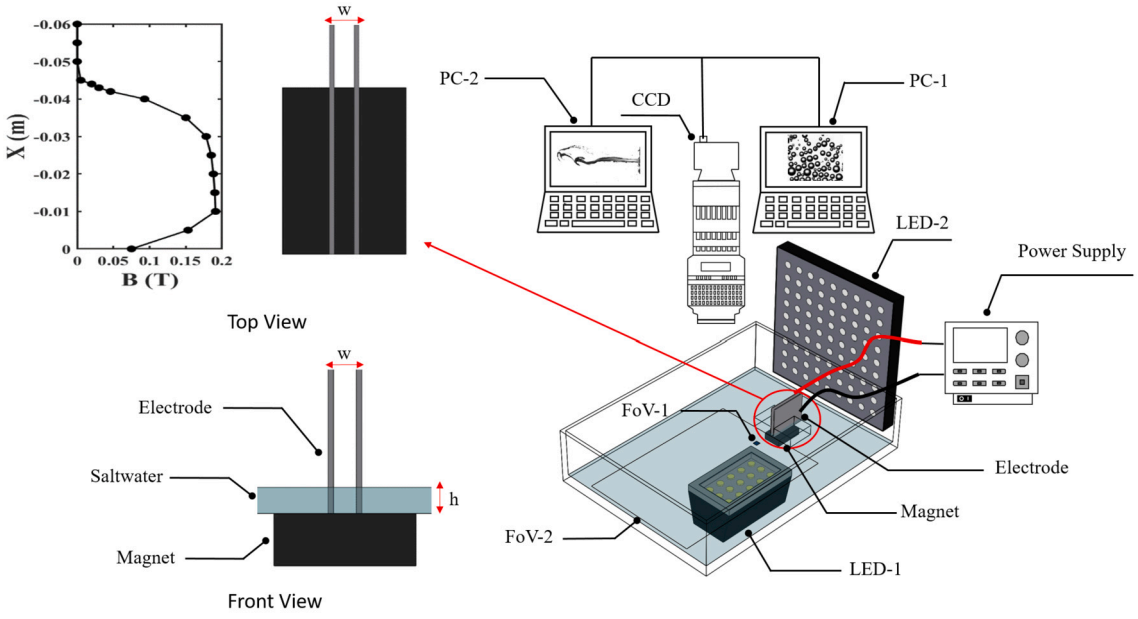


Fig. 1. Principle sketch of experimental apparatus.

In addition, the magnitude of magnetic field does not affect the bubble growth rate. To further improve the production efficiency, optimal configurations of magnetic field and electrodes are proposed [32].

On the other hand, conventional jet flow had been thoroughly investigated for decades as reviewed in Ref. [33]. The formation of turbulent jet is generally described as three main regions, i.e., the potential core, the developing region and the self-preservation region [34,35]. Also intensively studied is the two phase jet, e.g., liquid jet associated with atomizing gas stream [36,37]. A flapping instability is observed in an incompletely atomized liquid jet, in which regime of the flapping instability is determined by the wavelength and liquid jet diameter [36]. The MHD saltwater jet differs from the conventional liquid-gas jet by two features, i.e., driven by the Lorentz force and chemical reactions associated with electrolysis. While the Lorentz force can be used for marine vehicle propulsion, electrolysis may be potentially applied for hydrogen collection. To study the coupling effect of bubble generation by electrolysis with hydrodynamics, forced saltwater flow with electrolysis is experimented [38]. Bubble size is found to be controlled by the current density. Larger bubbles are observed for stronger current density or slower flow. Recent study of MHD bubbly jet found the flow transition from laminar to turbulent is determined by a modified Reynolds number based on the Lorentz force [39] defined as:

$$Re_L = \frac{\sqrt{\rho B_0 D_H I}}{\mu}, \tag{1.1}$$

where ρ, B_0, D_H, I and μ represent density, magnetic field strength, hydraulic diameter, current strength and viscosity, respectively.

Most of the reported MHD works of electrolyzed bubbles mainly focus on the application of enhanced hydrogen production, which the Lorentz force orients in line with buoyancy. On the other hand, size and motion of bubbles consisted in a bubbly MHD jet, in which the Lorentz force orients perpendicularly to buoyancy, are rarely studied. In addition, the MHD jet not just provides thrust needed for marine propulsion, but also produces bubbles of hydrogen and oxygen by associated electrolysis. These produced bubbles might possess further implications to practical applications. For instance, the hydrogen may be collected for possible source as supplemental energy in a hybrid power system. Alternatively, bubbles may be collected and guided to surround the sailing marine vehicle for drag reduction. Better understandings of the produced bubbles in a MHD bubbly jet are important to optimize these applications. In this paper, the coupling effect regarding the size and motion of electrolyzed bubbles in the Lorentz-force-driven MHD bubbly jet is presented by parametric analysis of experimental results.

2. Experimental apparatus

The experimental apparatus shown in Fig. 1 is mostly identical with the setup used in Ref. [39], except the material of electrodes. In the present study, a pair of titanium electrodes of length $l_e = 55$ mm are held by two acrylic plates on their sides with width w in between. The electrodes are placed inside a rectangular acrylic tank partially filled with saltwater of depth h. This configuration forms an open channel of saltwater. The saltwater used in the experiments is the solution of distilled water and sodium chloride. Density and viscosity of this saltwater solution is $\rho = 1023$ kg/m^3 and $\mu = 0.00097$ N s/m^2, respectively. The electric conductivity of the saltwater is $\sigma = 5$ S/m measured by conductivity meter, where S stands for Siemens. A permanent NdFeB magnet is placed right beneath the tank to provide an upward magnetic field **B**. The magnitude of magnetic field measured along the central line of the open channel is also shown in Fig. 1. By the present setup, a magnetohydrodynamic (MHD) jet is discharged from the open channel by the Lorentz force, whose force density **F** is expressed as

$$\mathbf{F} = \mathbf{J} \times \mathbf{B}. \tag{2.2}$$

Here, **J** represents the current density. The current density may be approximated as $J \approx I/A_w$, where A_w is the wetted area of electrode imposed in experiments. For the latter analysis, the average field strength of $B_0 = 0.147$ T is taken as a representative strength to obtain Re_L. Power suppliers are connected to generate an electric current **I** through the conductive saltwater. The induced Lorentz force is aligned with the axial direction of the open channel. It is noticed that, the current density is affected by the water depth h for fixed electrode length l_e in the present study, i.e., $A_w = hl_e$. In the meantime, saltwater undergoes electrolysis, so that hydrogen, chlorine and oxygen are generated. Several possible reactions can take place depending on the concentration of sodium chlorine. In the present condition of intermediate concentration, the reaction is expected as

$$4H_2O + 2NaCl \rightarrow 3H_2 + O_2 + Cl_2 + 2NaOH. \tag{2.3}$$

As a result, the gaseous bubbles include hydrogen, chlorine and oxygen. It is noticed that, chlorine gas will rapidly dissolve in the saltwater

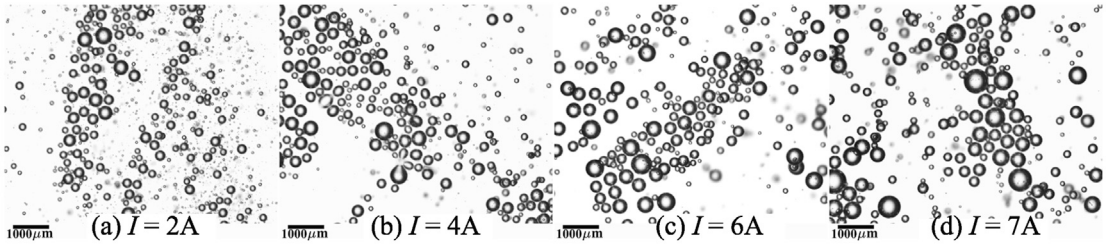


Fig. 2. Representative snapshot of bubbles in MHD bubbly jets with varied electric current I from a channel of $w = 4$ mm and $h = 5$ mm. Shown from left to right are (a) $I = 2$ A ($Re_L = 1192$), (b) $I = 4$ A ($Re_L = 1686$), (c) $I = 6$ A ($Re_L = 2064$) and (d) $I = 7$ A ($Re_L = 2230$). Larger bubbles are generally produced in higher I.

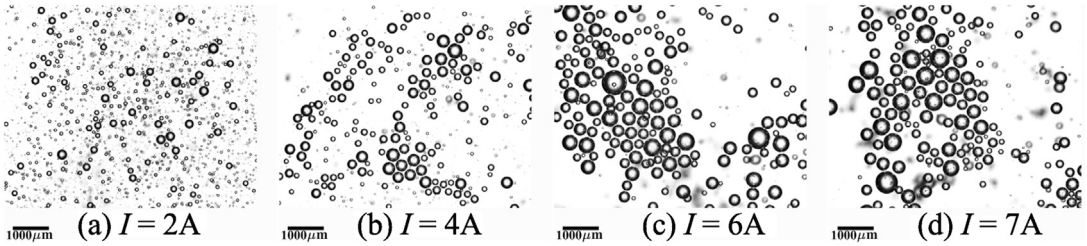


Fig. 3. Representative snapshot of bubbles in MHD bubbly jets with varied electric current I for $w = 4$ mm and $h = 10$ mm. Shown from left to right are (a) $I = 2$ A ($Re_L = 1352$), (b) $I = 4$ A ($Re_L = 1911$), (c) $I = 6$ A ($Re_L = 2341$) and (d) $I = 7$ A ($Re_L = 2528$).

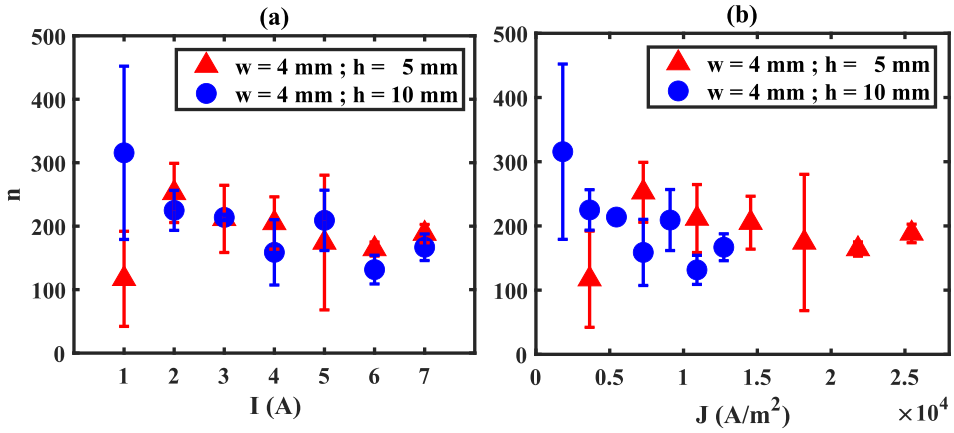


Fig. 4. Number of bubbles n per frame vs. (a) electric current I, and (b) current density J.

[2,38], so that the bubbles observed are mainly oxygen and hydrogen. Since the main interest is to understand the characteristics and motion of bubbles, bubbles are treated as the same gas component with a constant surface tension (γ) to the saltwater. To analyze the detailed characteristics of bubble and formation of the whole bubbly jet, two top views are recorded by CCD cameras, referred to as field of view 1 (FoV-1) and field of view 2 (FoV-2) shown in Fig. 1. Images of FoV-1, taken by a Mikrotron MotionBlitz camera with a frame speed of 100 fps, focus on the close-up view of bubbles. On the other hand, a GoPro Action (60 fps) is used to observe the full view of bubbly jet covered by the FoV-2. The recorded pictures are transmitted to computers PC-1 and PC-2, respectively, and imported into the Matlab software for further analysis. The main parameters analyzed include the mean diameter and the Sauter diameter of bubbles, denoted as D_m and D_S, respectively. To gain insights of bubble motion, diameter and speed of bubbles, denoted respectively as D_b and U_b, will also be measured and discussed.

3. Results and discussion

3.1. Distributions of bubble size

This section focuses on the characteristics of bubbles produced by electrolysis and transported by the Lorentz force. Shown in Fig. 2 are the representative images taken in FoV-1 for various current strengths $I = 2$ A, 4 A, 6 A and 7 A with a channel geometry of $w = 4$ mm and $h = 5$ mm. Bubbles of various sizes are clearly observed. Based on visual inspection, the overall size of bubbles tends to increase as the current strength is raised higher. Similar trend is also observed for the cases in a deeper height of $h = 10$ mm as shown in Fig. 3. Nevertheless, the bubbles for $I = 2$ A and 4 A in this deeper channel appear smaller than the correspondent cases of identical current strength shown in Fig. 2. It is noticed that, the sizes of bubbles are visibly comparable for cases of $I = 2$ A shown in Fig. 2 and $I = 4$ A shown in Fig. 3, whose current densities J are nearly identical. This indicates that bubble size is more relevant to J which directly governs the Lorentz force, rather than I. Quantitative verifications of these qualitative observations will be presented in latter sections.

The first quantitative measure interested is the numbers of bubbles. Shown in Fig. 4 is the average bubble count n within the FoV-1 taken between 5 s < t < 30 s (2500 frames in total) for various current strengths and current densities. In general, n does not vary significantly, e.g., $n \approx 200$ as shown in Fig. 4, except for the cases of very low I (or J) when the Lorentz-induced flow is weak. Similar trends regarding the size and number of bubbles are observed for a wider $w = 6$ mm associated with various h and I, e.g., $h = 5$ mm and 10 mm, $I = 2$, 4 A, 6 A and 7 A. The representative snapshots are shown in Fig. 5.

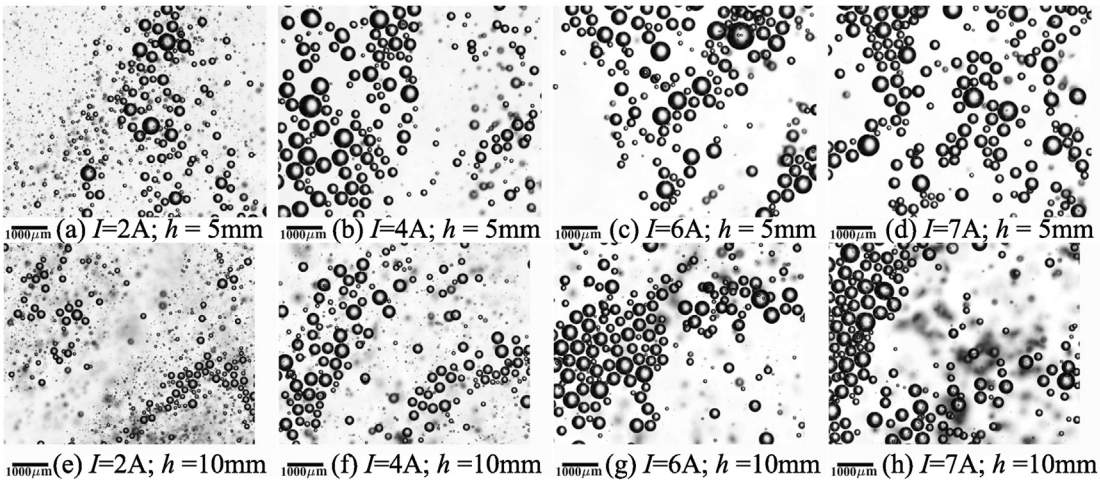


Fig. 5. Representative snapshot of MHD bubbly jets for varied electric current I for $w = 6$ mm. Top row: $h = 5$ mm, (a) $I = 2$ A ($Re_L = 1320$), (b) $I = 4$ A ($Re_L = 1867$), (c) $I = 6$ A ($Re_L = 2287$) and (d) $I = 7$ A ($Re_L = 2470$). Bottom row: $h = 10$ mm, (e) $I = 2$ A ($Re_L = 1548$), (f) $I = 4$ A ($Re_L = 2190$), (g) $I = 6$ A ($Re_L = 2682$) and (h) $I = 7$ A ($Re_L = 2897$).

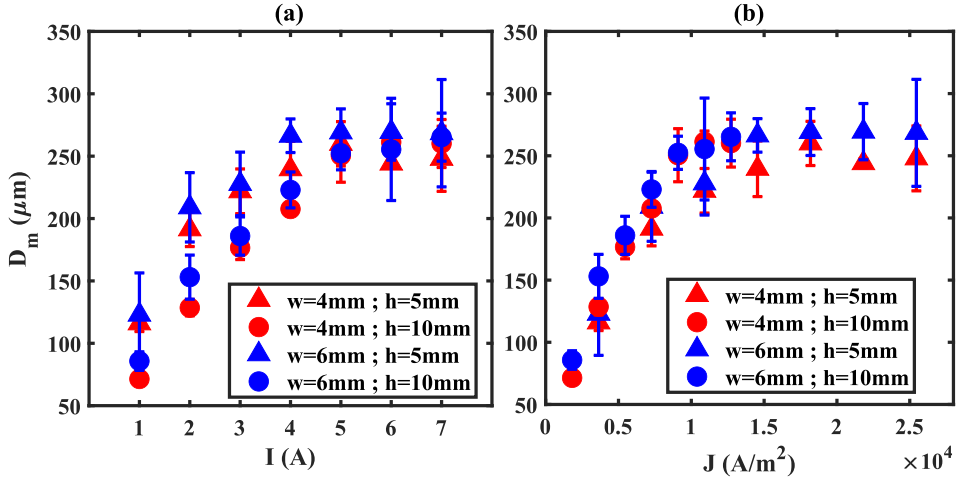


Fig. 6. Mean Diameter of bubbles D_m vs. (a) electric current I, and (b) current density J. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Size of bubbles is one of the main concerns when investigating the bubble dynamics. The mean diameter of bubbles D_m is measured between 25 s < t < 30 s (500 frames) from three different experiments under identical conditions. Shown in Fig. 6 are the mean diameters in various current strength and current density. Sizes of bubbles under the present experimental conditions are in $O(100)$ μm. Since the width w does not affect the overall current density, so that its effect to D_m is insignificant. D_m increases as I is raised higher as shown in Fig. 6(a). Nevertheless, the increase of bubble size seems less dramatic for higher current strength. Also observed in the range of weaker current strength $I \leq 4$ A, the bubbles appear apparently smaller for a deeper channel of $h = 10$ mm. On the other hand, the mean diameter D_m can be nicely described for various experimental conditions if plotted versus current density J as shown in Fig. 6(b). Two distinct regimes can be clearly distinguished, such that: an apparent increase of bubble size for $J < 10^4$ A/m^2 until asymptotically reaching to nearly constant size for $J > 10^4$ A/m^2. The asymptotic diameter $D_m \approx 270$ μm is in line with the early findings even the configuration of MHD effect is not identical [30]. Similar behavior is also observed for the Sauter diameter D_S as shown in Fig. 7. It is noticed that, even the trend of larger bubbles generated by stronger current density has been well reported previously [29,30,38], those early experiments do not vary the area of electrodes to verify the general dependence shown in Fig. 6(b). In addition, existence of the distinct regimes suggests different dominant mechanisms within these two regimes, hereinafter referred to as (1) the electrochemistry regime and (2) the flow regime, respectively, as the reasons addressed in the latter sections.

It has been reported that, size of bubble appears larger for either stronger current density or slower forced flow speed [38]. In the present study, flow is induced by the Lorentz force, so that the speed is also proportional to the current density. This confirms the bubble size can be well described by the current density as shown in Figs. 6(b) and Fig. 7(b). Nevertheless, two distinct regimes, i.e., the electrochemistry regime and the flow regime, are identified. Within the electrochemistry regime of $J < 10^4$ A/m^2, bubble size increases as current density is raised higher. On the other hand, the bubble size maintains nearly unchanged in the flow regime, once the current density exceeds 10^4 A/m^2. The reason of observed regime transition of bubble size is discussed in the next section.

3.2. Dynamics of bubble motion

It is understood that an equilibrium bubble size (D_b) exists by balancing all the surface forces, including the surface tension, the static hydrodynamic pressure, the pressure inside the bubble (p_{gas}) and the ambient pressure (p_{atm}), expressed as:

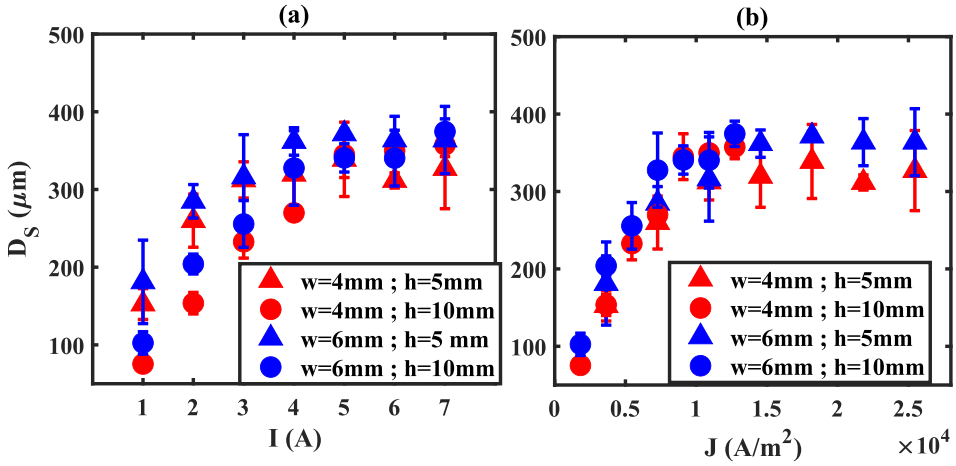


Fig. 7. Sauter Diameter of bubbles D_S vs. (a) electric current I, and (b) current density J.

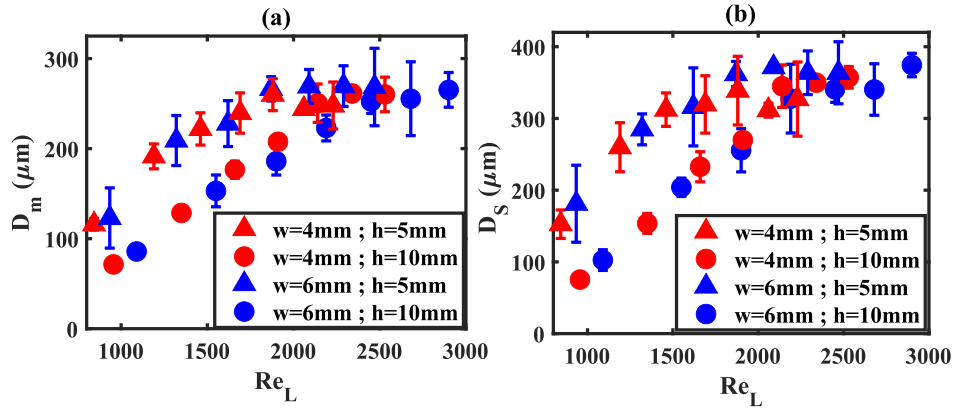


Fig. 8. Bubble sizes for various Reynolds numbers Re_L: (a) mean diameter D_m, and (b) Sauter diameter D_S.

$$p_{gas} + \frac{2\gamma}{D_b} = p_{atm} + \rho g h. \tag{3.4}$$

The surface tension between air and water at 25^oC and the typical equilibrium bubble diameter taken from Fig. 6(b) with $h = 5$ mm are $\gamma \approx 72$ mN/m and $D_b \approx 270$ μm, respectively. As a result, the pressure difference inside bubble and ambience is estimated as $p_{atm} - p_{gas} \approx 500$ Pa. The pressure inside bubble is mainly determined by the resultant gas of electrolysis. Consequently, the size of bubble would be affected by the current, which directly dominates the gas generation.

The flow pattern of jet, e.g., laminar or turbulent, can be determined by the Reynolds number based on the Lorentz force [39] Re_L. To understand how the flow induced by the Lorentz force affects the formation of bubbles, the mean bubble sizes D_m and D_S are plotted versus the corresponding Re_L as shown in Fig. 8. Again, distinct relations of the bubble sizes to the Re_L are observed. For $Re_L < 2000$ in Fig. 8, even the bubble size consistently increases with Re_L, the actual size is smaller for deeper $h = 10$ mm, or weaker current density for a fixed current strength. This indicates that, effect of current density to the bubble size is important during this range of Re_L. This range corresponds the electrochemistry regime proposed in the previous sections. Considering the cases of pure electrolysis ($Re_L = 0$), the bubble sizes would be still varied for different current density. As a result, even the growth rate against Re_L is similar, the actual bubble size is still strongly affected by J. It indicates bubble size in this regime can not be solely described by Re_L, so that the flow is not the dominant factor. On the contrast, once $Re_L > 2000$, the bubble sizes appear nearly the same for all conditions regardless the depth of channel h. In this regime, the flow parameter of Re_L is capable to solely describe the bubble size, so that it is referred to as the flow regime.

To better understand the dynamics of bubbles, motions of bubbles, represented by several randomly selected bubbles, are measured. Shown in Fig. 9 are 3 consecutive snapshots of two relatively weaker current strengths of $I = 2$ A and 4A. Average speeds of the two representative bubbles are calculated in the order of $O(10^{-1})$ m/s, and also shown in the figures. As expected, bubbles move faster for higher current strength. This trend is preserved for stronger current strengths. It is noteworthy that, bubbles frequently proceed burst and coalescence under stronger current strength. The emergence of these distinct behaviors also suggests an important flow transition triggered in the cases associated with sufficiently strong current strength or high Re_L.

The average speeds of numerous representative bubbles U_b for various conditions are shown in Fig. 10. Even a trend of monotonically higher U_b with increasing J is observed as shown in Fig. 10(a), U_b appears not solely dependent on J, but also the channel width w (or the hydraulic diameter of channel). To analyze motion of bubbles, the Reynolds number Re_b based on D_b and U_b of individual bubble is defined as $Re_b = \rho U_b D_b/\mu$. It should be noticed that, this Re_b is to represent the motion of disperse phase of bubbles, and should not be confused with the Re_L, which represents the continuous phase of carrier saltwater. In other words, the flow pattern is determined by Re_L but Re_b.

The Re_b versus the corresponding experimental condition of Re_L is plotted in Fig. 10(b). A consistent trend of monotonically higher Re_b resulting in larger Re_L is obtained. Again, the growth rate of Re_b is

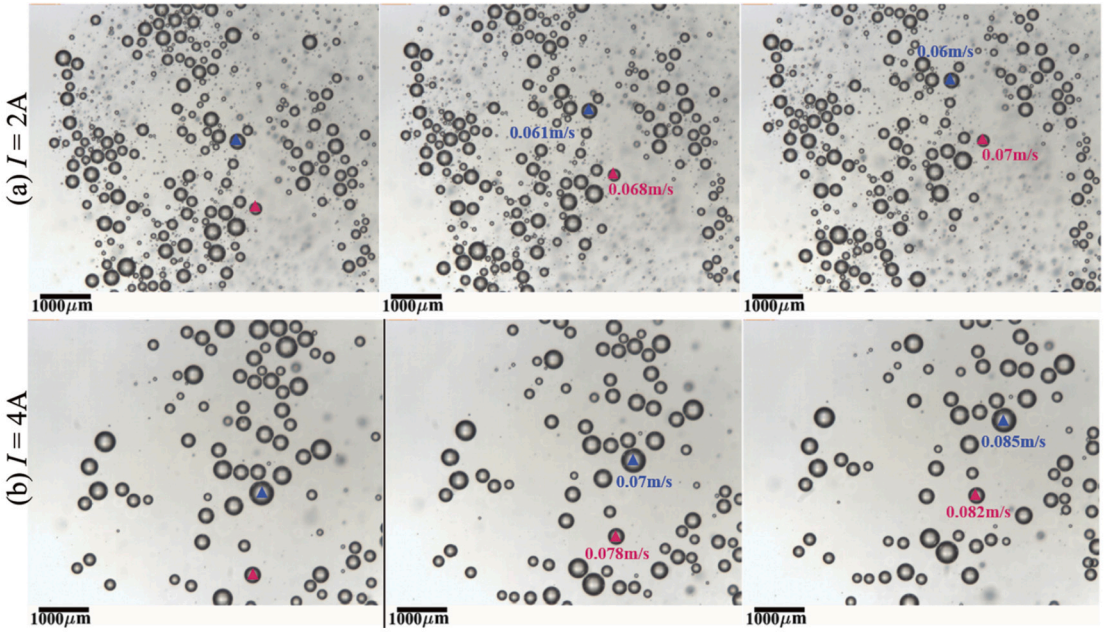


Fig. 9. Bubble motion: time sequence from left to right. Speeds of two selected bubbles, marked by triangle, are shown. (a) top row: $w = 4$ mm, $h = 5$ mm and $I = 2$ A; (b) bottom row: $w = 4$ mm, $h = 5$ mm and $I = 4$ A.

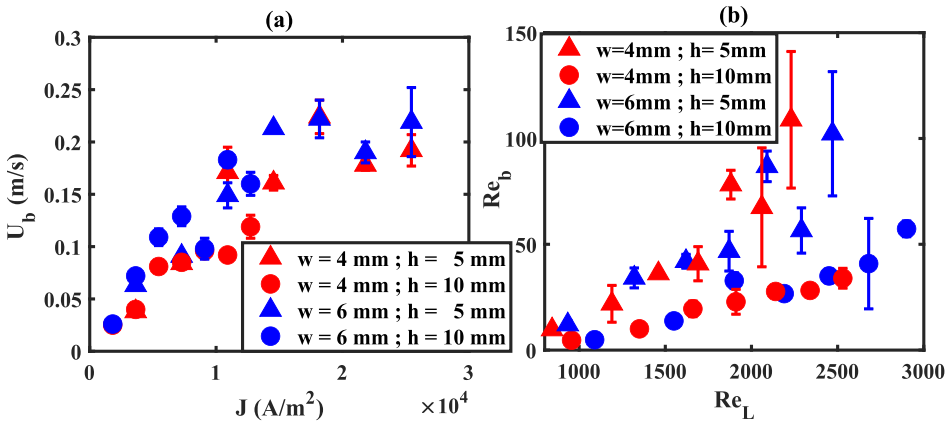


Fig. 10. (a) Speed of selected bubbles U_b for various current density J. (b) Reynolds number of selected bubbles Re_b vs. corresponding flow Reynolds number Re_L.

larger for a smaller h (or higher J). The consistent correlation between the two Reynolds numbers confirms that, the hydrodynamics prescribed by Re_L is capable to describe the trend of bubble motion represented by Re_b. Nevertheless, the distinct quantitative dependence of the two Reynolds numbers on h (or J) indicates another important factor of electric effect.

The results presented above, e.g., D_m and D_S, indicate a transition in $Re_L > 2{,}000$. To elucidate the flow transition, multiple experiments are carried out for every presented case to observe the corresponding pattern of bubbly jet. The representative images of bubbly jets for various Re_L taken by FoV-2 are shown in Fig. 11. Even the downstream jet front appears apparent interfacial instability for $Re_L = 1{,}351$, formation of core region near the channel exit is well preserved, which can be qualitatively defined as a laminar jet. The downstream jet front starts to breakup for $Re_L = 1{,}911$. The transition of pattern agrees with the previous findings of MHD bubbly jet with different electrodes [39], where more details of the flow patterns can be referred. Significant turbulence breakup evolves further upstream to the core region of the jet for $Re_L \geq 2{,}137$. It is interesting to notice that, the Re_L of transition from the electrochemistry regime to the flow regime shown in Fig. 8 is consistent with the apparent breakup of jet core. This consistency suggests that the bubble size is mainly dominated by the breakups of jet core for the flow regime of $Re_L > 2{,}000$, where local turbulent flow universally enhances the detachments of bubbles from electrode plates regardless the magnitude of applied electrical field.

4. Conclusion

Magnetohydrodynamic (MHD) jet driven by the Lorentz force is relevant to the potential applications, such as marine propulsion and hydrogen production. The present experimental study analyzes the characteristics of MHD bubbly jet, such as the size distributions of bubbles produced in electrolysis. Under a fixed magnetic field at about $B_0 \approx 0.147$ T associated with current density $J \leq 2.5 \times 10^4$ A/m^2, the size of gaseous bubbles of oxygen and hydrogen is determined by the current density and ranges from $O(10)$ μm ~ $O(100)$ μm. For condition associated with relatively weaker current density, mean bubble size is larger as the current density is increased. However, once the current density reaches sufficiently high, e.g., $J \approx 10^4$ A/m^2, the size remains

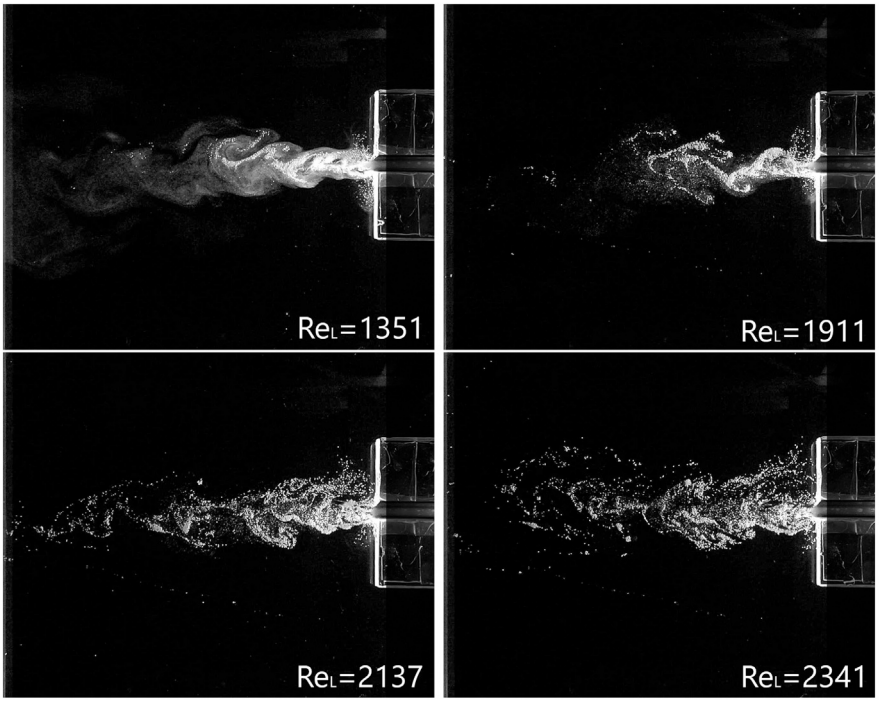


Fig. 11. Full view of MHD bubbly jet for various Reynolds numbers.

nearly constant. The distinct dependences of size on the current density suggest different dominant mechanisms in these two regions.

To elucidate the coupling effect of the Lorentz force and the electrolysis to the size of bubble, mean diameter of bubbles versus a modified Reynolds number Re_L is analyzed. Similar behavior is observed, such that larger bubbles are produced for higher Re_L until maximum bubble size is reached for $Re_L > 2,000$. Nevertheless, even bubble size grows for increasing Re_L for $Re_L \leq 2,000$, the size appears different for various current density. The size difference indicates the mean bubble diameter can not be solely described by the flow pattern in this region of Re_L. Dynamics of bubble motion are also investigated, which is presented by the bubble-based Reynolds number Re_b. In addition to higher bubble speed (or Re_b) for increasing Re_L as expected, merges and bursts of bubbles are often observed under sufficiently high Re_L.

Inability of Re_L to universally correlate the bubble size suggests the distribution of bubble size is not fully affected by hydrodynamics in the present MHD flow. Two regimes, referred to as the electrochemistry regime and the flow regime, are categorized. For condition of relative weaker magnetoelectric effect, i.e., smaller current density or $Re_L \leq 2000$, size of bubble increases with the magnitude of current density. The electrochemistry regime is referred for this significant dependence between bubble size and current density. Nevertheless, for sufficiently strong magnetoelectric effect (high current density or $Re_L > 2000$), breakup of jet core is triggered by turbulence. The mean size of bubbles remains nearly unchanged regardless the magnitude of applied electric field. Since the constant bubble size is mainly affected by the flow transition of turbulent breakups, so that referred to as the flow regime. The flow transition is supported by the corresponding pattern of bubbly jet.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Alfven H. Existence of electromagnetic-hydrodynamic waves. Nature 1942;150:450.
[2] Cebron D, Viroulet S, Vidal J, Masson JP, Viroulet P. Experimental and theoretical study of magnetohydrodynamic ship models. PLoS ONE 2016;12:e0178599.
[3] Chan CS, Cheng JH, Zeng CH, Huang JR, Chen YH, Chen YJ, et al. Design of marine vehicle powered by magnetohydrodynamic thruster. Magnetohydrodynamics 2020;56:51.
[4] Jang J, Lee S. Theoretical and experimental study of MHD (magnetohydrodynamic) micropump. Sens Actuators A, Phys 2000;80:84.
[5] Timofeev VN, Khatsayuk MY, Kizhaev IV. Mathematical simulation of electromagnetic and hydrodynamic processes in the MHD pump. Magnetohydrodynamics 2019;55:337.
[6] Haghparast M, Pahlavani M, Azizi D. Numerical investigation of the effects of magnetic field and fluid electrical conductivity on the performance of marine magnetohydrodynamic motors. IET Electr Power Appl 2018;12:1207.
[7] Lin CH, Chen PW, Chen CY. Simulations of silicon CZ growth in a cusp magnetic field. Magnetohydrodynamics 2011;47:17.
[8] Chen JC, Chiang PY, Nguyen T, Hu C, Chen CH, Liu C. Numerical simulation of the oxygen concentration distribution in silicon melt for different crystal lengths during Czochralski growth with a transverse magnetic field. J Cryst Growth 2016;452:6–11.
[9] Selimefendigil F, Oztop HF. Mixed convection of nanofluid filled cavity with oscillating lid under the influence of an inclined magnetic field. J Taiwan Inst Chem Eng 2016;63:202–15.
[10] Ghadikolaei SS, Gholinia M, Hoseini ME, Ganji DD. Natural convection MHD flow due to MoS_2-Ag nanoparticles suspended in $C_2H_6O_2$-H_2O hybrid base fluid with thermal radiation. J Taiwan Inst Chem Eng 2019;97:12–23.
[11] Hashim, Hamid A, Khan M. Multiple solutions for MHD transient flow of Williamson nanofluids with convective heat transport. J Taiwan Inst Chem Eng 2019;3:26–137.
[12] Alizadeha R, Mesgarpourb M, Americ A, Abadd J, Wongwisesb S. Artificial intelligence prediction of natural convection of heat in an oscillating cavity filled by CuO nanofluid. J Taiwan Inst Chem Eng 2021;124:75–90.
[13] Zhanga L, Bhattia MM, Shahidb A, Ellahic R, Begd O, Saite SM. Nonlinear nanofluid fluid flow under the consequences of Lorentz forces and Arrhenius kinetics through a permeable surface: a robust spectral approach. J Taiwan Inst Chem Eng 2021;124:98–105.
[14] Yuan MH, Chang CC, Chang CY, Lin YY, Shie JL, Wu CH, et al. Radio-frequency-powered atmospheric pressure plasma jet for the destruction of binary mixture of naphthalene and n-butanol with Pt/Al_2O_3 catalyst. J Taiwan Inst Chem Eng 2014;45:468–74.

[15] Sar A, Shabani E, Haghighi M, Shabani M. Synergistic catalytic degradation of ciprofloxacin using magnetic carbon nanomaterial/$NiFe_2O_4$ promoted cold atmospheric pressure plasma jet: influence of charcoal, multi walled carbon nanotubes and walnut shell. J Taiwan Inst Chem Eng 2022;132:104131.
[16] Ball M, Wietschel M. The future of hydrogen-opportunities and challenges. Int J Hydrog Energy 2009;34:615–62.
[17] Dincer I. Green methods for hydrogen production. Int J Hydrog Energy 2012;37:1954–71.
[18] Sivanantham A, Ganesan P, Shanmugam S. Hierarchical $NiCo_2S_4$ nanowire arrays supported on Ni foam: an efficient and durable bifunctional electrocatalyst for oxygen and hydrogen evolution reactions. Adv Funct Mater 2016;26:4661–72.
[19] Badwal SPS, Giddey S, Munnings C. Hydrogen production via solid electrolytic routes. Wiley Interdiscip Rev Energy Environ 2013;2:473–87.
[20] Nikolic VM, Tasic GS, Maksic AD, Saponjic DP, Miulovic SM, Marceta KMP. Raising efficiency of hydrogen generation from alkaline water electrolysis-energy saving. Int J Hydrog Energy 2010;35:12369–73.
[21] Koza JA, Muhlenhoff S, Zabinski P, Nikrityuk PA, Eckert K, Uhlemann M, et al. Hydrogen evolution under the influence of a magnetic field. Electrochim Acta 2011;56:2665–75.
[22] Lin MY, Hourng LW, Kuo CW. The effect of magnetic on hydrogen production efficiency in water electrolysis. Int J Hydrog Energy 2012;37:1311–20.
[23] Chen YJ, Li YH, Chen CY. Studying the effect of electrode material and magnetic field on hydrogen production efficiency. Magnetochemistry 2022;8:53.
[24] Wang M, Wang Z, Gong X, Guo Z. The intensification technologies to water electrolysis for hydrogen production-a review. Renew Sustain Energy Rev 2014;29:573–88.
[25] Zhang Y, Liang C, Wu J, Liu H, Zhang B, Jiang Z, et al. Recent advances in magnetic field-enhanced electrocatalysis. ACS Appl Energy Mater 2020;3:10303–16.
[26] Matsushimaa H, Iida T, Fukunakab Y. Gas bubble evolution on transparent electrode during water electrolysis in a magnetic field. Electrochim Acta 2013;100:261–4.
[27] Chandran P, Bakshi S, Chatterjee D. Study on the characteristics of hydrogen bubble formation and its transport during electrolysis of water. Chem Eng Sci 2015;138:99–109.
[28] Lin MY, Hourng LW, Wu CH. The effectiveness of a magnetic field in increasing hydrogen production by water electrolysis. Energy Sources, Part A, Recovery Utilization Environ Eff 2017;39:140–7.
[29] Li Y, Yang G, Yu S, Kang Z, Mo J, Han B, et al. In-situ investigation and modeling of electrochemical reactions with simultaneous oxygen and hydrogen microbubble evolutions in water electrolysis. Int J Hydrog Energy 2019;44:28283–93.
[30] Zhan S, Huang Y, Zhang W, Li B, Jiang M, Wang Z, et al. Experimental investigation on bubble growth and detachment characteristics on vertical microelectrode surface under electrode-normal magnetic field in water electrolysis. Int J Hydrog Energy 2021;46:36640–51.
[31] Mohammadpour H, Cord-Ruwisch R, Pivrikas A, Ho G. Utilisation of oxygen from water electrolysis-assessment for wastewater treatment and aquaculture. Chem Eng Sci 2021;246:117008.
[32] Li YH, Chen YJ. The effect of magnetic field on the dynamics of gas bubbles in water electrolysis. Sci Rep 2021;11:9346.
[33] Lasheras JC, Hopfinger EJ. Liquid jet instability and atomization in a coaxial gas stream. Annu Rev Fluid Mech 2000;32:275.
[34] Pearce AF. Critical Reynolds number for fully-developed turbulence in circular submerged water jets. Council for Scientific and Industrial Research, Report MEG 475, Pretoria, South Africa. August 1966.
[35] Magstadt AS. Investigating the structures of turbulence in a multi-stream, rectangular, supersonic jet. Ph.D. Dissertation. Syracuse University; June 2017.
[36] Delon A, Cartellier A. Flapping instability of a liquid jet. Phys Rev Fluids 2018;3:043901.
[37] Xiong P, He S, Qiu F, Cheng Z, Quan X, Zhang X, et al. Experimental and mathematical study on jet atomization and flash evaporation characteristics of droplets in a depressurized environment. J Taiwan Inst Chem Eng 2021;123:185–98.
[38] Boissonneau P, Thibault J-P. Experimental analysis of couplings between electrolysis and hydrodynamics in the context of MHD in seawater. J Phys D, Appl Phys 1999;32:2387–98.
[39] Cheng JH, Cheng LW, Chen CY. Flow transition of magnetohydrodynamic bubbly jet driven by Lorentz force. J Taiwan Inst Chem Eng 2022;135:104369.